Proof of Concept
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An In Silico Self-Calibrating Simulator for Caffeine Dosing in Apnea of Prematurity

Learning Hidden Patterns from Clinical Data

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Abstract

Apnea of prematurity (AOP), a cessation of breathing, affects nearly all preterm infants born before 29 weeks' gestation and is caused mainly by immaturity of brainstem respiratory control, sometimes combined with airway obstruction. Apnea episodes are accompanied by hypoxemia (an abnormally low level of oxygen) and bradycardia (a slower-than-normal heart rate). While occasional apneas may be normal, a higher burden can be harmful and is associated with long-term neurodevelopmental impairment. As a first-line pharmacological treatment, caffeine citrate has been used for decades as a respiratory stimulant. Doses of caffeine citrate are typically based on weight and postnatal age, following a one-size-fits-all regimen, and rigid therapeutic ranges are assumed to apply to all infants. However, infants' responses to the drug differ considerably beyond weight and age: preterm infants vary in their sensitivity to caffeine, their level of maturity and their tolerance of high concentrations. Moreover, the same infant changes over time, as caffeine clearance rises about fourfold during the first month of life, so a fixed dose delivers less and less exposure.

Consequently, both under- and over-dosed infants are exposed to avoidable risks that a dose tailored to each infant's needs could prevent. Therapeutic drug monitoring, which measures caffeine in the blood, can reveal how much drug an infant actually carries, but it requires repeated blood sampling in infants with very small blood volumes and is therefore not routinely recommended. Furthermore, there are currently no validated biomarkers to guide treatment or dosing, only a limited number of datasets on caffeine dosing for AOP, and more than 80% of the apneas identified on vital sign recordings are missed in clinical notes. An individualized, non-invasive dosing strategy is therefore needed to reduce the risk of adverse effects while optimizing the efficacy of the drug. Several studies propose candidate biomarkers of caffeine response, such as the brain age gap, since AOP reflects the immaturity of the nervous system.

We propose an in silico self-calibrating simulator for individualized caffeine dosing in AOP using only non-invasive data, with a major focus on infants who do not respond to standard regimens. Each infant's hidden parameters start from population priors built from published pharmacokinetic models and the infant's covariates, and are updated daily from the observed apnea count and the doses actually given during the treatment period (which can last from weeks to months), while candidate biomarkers are treated as distributions refined with clinical data. The system can therefore learn the infant's hidden parameters, such as the caffeine concentration in the blood, and suggest an individualized dose. The system is strictly advisory: it suggests a dose for clinician review and keeps learning from whatever dose is given. It is intended for research and preclinical testing of dosing strategies.

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Abbreviations: (AOP) apnea of prematurity, (BW) birth weight, (CL) clearance, (ECG) electrocardiography, (EEG) electroencephalography, (GA) gestational age, (HR) heart rate, (IBI) inter-breath interval, (IH) intermittent hypoxemia, (IIV) inter-individual variability, (IOV) inter-occasion variability, (NICU) neonatal intensive care unit, (PK) pharmacokinetics, (PMA) postmenstrual age, (PNA) postnatal age, (SpO_2) peripheral oxygen saturation, (TDM) therapeutic drug monitoring, (V_d) volume of distribution

Keywords: apnea of prematurity, caffeine, computer simulation, Bayesian calibration, biomarkers, neonatal intensive care unit

1 PURPOSE OF THE STUDY

Apnea of prematurity (AOP) is defined as a pause in breathing lasting more than 20 s, or a shorter pause accompanied by bradycardia, cyanosis, or oxygen desaturation, in an infant born before 37 weeks' gestation [1, 2]. Its incidence rises steeply as gestational age falls: AOP affects nearly all infants born at <29 weeks' gestation or weighing <1,000 g, compared with 54% of infants born at 30–31 weeks, 15% at 32–33 weeks, and 7% at 34–35 weeks [3].

Caffeine citrate is the most widely used drug for AOP. It reduces bronchopulmonary dysplasia (a chronic lung disease that primarily affects premature infants who require prolonged oxygen therapy or mechanical ventilation) and shortens the duration of positive airway pressure [4], and improves survival without neurodevelopmental disability at 18–21 months' corrected age [5]. It is dosed by protocol on the basis of body weight and postnatal age. The World Health Organization recommends a loading dose of 20 mg/kg of caffeine citrate followed by a maintenance dose of 5 mg/kg/day [6]. After the CAP trial, many centers adopted maintenance doses of 5–10 mg/kg/day [2].

However, recent reviews call for individualized dosing guided by patient characteristics [2, 7], as such fixed regimens are unlikely to suit every infant, especially since some infants do not respond even when caffeine levels are in the "therapeutic" range (5–20 mg/L) [8], while others reach suggested toxic levels on standard maintenance doses [9]. Because infants differ in their sensitivity to the drug, their optimal dose may lie above or below the standard regimen. A fixed regimen also fails the same infant over time: caffeine clearance rises about fourfold in the first month [10]. Dosing must therefore be guided to avoid toxicity (risk of tachycardia, agitation and vomiting and, at the extreme, cerebellar (brain) hemorrhage and seizures [11]), especially with high-dose therapy, suspected toxicity, treatment failure, or unusual pharmacokinetics.

In silico simulation offers a way around this. The UVA/Padova Type 1 Diabetes Simulator, accepted by the FDA in 2008 as a substitute for animal trials in the preclinical testing of insulin-dosing algorithms [12], showed that a validated virtual population can serve as a credible testbed for dosing algorithms. Inspired by that work, we propose an in silico self-calibrating simulator that models the pharmacokinetics of caffeine in preterm infants. It allows different dosing strategies, in particular adaptive and individualized ones, to be evaluated on a virtual population in terms of their effect on apnea events and adverse effects, providing a platform for stress-testing dosing strategies before they are considered for clinical evaluation.

The long-term goal is a research platform that supports individualized dosing for infants who do not respond to standard caffeine therapy, and a tool for the identification of suitable pharmacodynamic biomarkers for AOP.

2 PRELIMINARY LITERATURE REVIEW

2.1 Current Evidence and Gaps

Caffeine is the first-line treatment for AOP [4, 5], yet guidelines and units differ in initiation criteria, maintenance dose (5–20 mg/kg/day) and discontinuation, and about 10% of infants must restart it for recurring apnea [2]. Population PK models agree that clearance rises about fourfold over the first month but differ in covariates and variability [13, 14, 15, 10]. An external evaluation of twelve published caffeine models found that none met all predictive criteria in an independent cohort, while Bayesian updating with even one observation substantially improved predictions [16]. Model-based dose adjustment has targeted concentration rather than clinical outcome [17]. Concentration–effect evidence is sparse and conflicting: some infants have apneas at therapeutic levels [8, 18], while higher doses risk cerebellar hemorrhage and seizures [19, 11], and adenosine receptor polymorphisms may contribute [20, 21]. Therapeutic drug monitoring can guide dosing, but it requires repeated blood sampling in infants with very small blood volumes and is not routinely recommended [8, 22]. Candidate biomarkers such as monitor-derived apnea rate, intermittent hypoxemia and EEG brain age remain unvalidated [2, 23, 24], and routine records are a poor substitute: more than 80% of monitor-detected apneas are missed in clinical notes [2]. No AOP simulator links dose, concentration, maturation, apnea and biomarkers; given the uncertain evidence, such a platform should represent parameters as updatable distributions [25], as proposed here.

3 OBJECTIVES OF THE STUDY

The proposed system is an in silico self-calibrating simulator that models the pharmacokinetics of caffeine in preterm infants, its effect on apnea and toxicity, and the infant's maturation over time. It is intended for the in silico exploration of caffeine dosing strategies for apnea of prematurity, with particular focus on infants who respond poorly to standard dosing, and as a research tool for further understanding of the pharmacology of caffeine and the identification of pharmacodynamic biomarkers of AOP.

The system is non-invasive, requiring no additional blood sampling, and strictly advisory: it suggests a dose for clinician review and never administers one.

4 PROPOSED METHODOLOGY

4.1 Design Overview

This paper proposes a simulator of virtual preterm infants for caffeine dosing studies, capable of calibrating its parameters to each individual infant. Starting from priors set by the infant's covariates (calculated from the population models), the system updates its estimate of that infant's hidden parameters once a day, using the observed daily apnea count and the doses given. Three phases are therefore proposed that build on each other: Patient Initialization, Calibration, and Dose Recommendation.

Because caffeine is administered daily until apnea resolves, which can take months, the system runs in a closed loop over a predefined treatment horizon.

In the first phase, Patient Initialization, the system computes each infant's initial state (clearance, volume of distribution, elimination rate, half-life, steepness) from population models, given its covariates (gestational age, postmenstrual age, birth weight and sex). The second phase, Calibration, updates these priors once a day using the observed daily apnea count and its effects after the daily dosing. The model predicts the expected number of apneas from the doses given and the infant's current parameter estimates, then compares it with the observed daily apnea count through a count likelihood that allows for clustering of events. The population priors keep the estimates anchored when data are scarce. Because the update uses whatever dose was actually given, whether or not it followed the recommendation, the system keeps learning from routine care without requiring additional blood sampling. For this step, a Bayesian approach is proposed to update the distributions of the hidden parameters, while keeping the population priors as a reference. This is an inverse problem, where the system infers the hidden parameters from the observed data. The third phase, Dose Recommendation, turns the updated estimates into a suggested dose for clinician review.

4.2 Design Principles

1. **Every variable is sourced.** The system is designed to support further exploration of caffeine PK, PD and potential biomarkers, and to measure their effect. Moreover, no model, parameter, or biomarker enters the simulator without at least one published source that argues for its effects.
2. **Interpretability.** The system must explore different learning strategies and must be validated against clinical data from multiple centers, to ensure that it applies equally to every infant and that its outputs remain interpretable and open to review by clinicians.

5 LIMITATIONS

- **Validation.** The simulator will require extensive validation in order to be accepted.
- **Model uncertainty.** Learning hidden states introduces uncertainty in the parameters, since the simulator cannot represent the complete structure of the preterm infant.
- **Identifiability.** When only apnea events are observed, clearance, volume and caffeine sensitivity cannot be separated: the data identify only how fast the effect fades and the dose rate needed for a given effect. Individual dose-response can be learned from apnea data alone, but the infant's actual concentration remains uncertain unless an exposure measurement, such as therapeutic drug monitoring or saliva, is available.
- **Clinical data.** Available datasets are few and small, and routine records are incomplete, as most apneas are not charted. Dose changes made in response to recorded apneas can be accounted for in calibration, but changes driven by unrecorded events (infection, transfusion, extubation) can bias it.
- **Biomarkers.** Most candidate biomarkers are not routinely measured and have not been validated for caffeine response, so their effects may remain uncertain after calibration if the data are not sufficient. Caffeine itself increases EEG continuity, so brain-age markers recorded during treatment may partly reflect the drug rather than maturity.

6 CONCLUSIONS

In summary, we propose that an in silico simulator built from published pharmacokinetic models and candidate biomarkers may provide a non-invasive, systematic way to individualize caffeine dosing in AOP, exploring how the dose should evolve with the infant's maturation.

Further work should validate the simulator against clinical data from multiple centers, examine whether it applies equally to every infant, and make sure that its outputs remain interpretable and open to review by clinicians. Only after such validation could it advance, through prospective studies and clinical trials, toward informing dosing decisions at the bedside.

6.1 Future Vision

Once the simulator is validated and reviewed by clinicians, it can be extended through federated learning. Because neonatal data are highly sensitive, data sharing between hospitals is restricted by privacy laws. Federated learning addresses this through a siloed approach, where each hospital calibrates the simulator locally on its own data and shares only the updated parameter dis-

tributions, so the model is enriched across sites without any raw data leaving the hospital.

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